

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 6: Gyroscopic Principles & Basic Gyro Instruments

Syllabus Source: CAE Oxford (pp. 189-228) | Pass Target: $\geq 90\%$ AviationExam

1. FUNDAMENTAL PROPERTIES OF GYROSCOPES

A mechanical gyroscope consists of a heavy symmetrical rotor spinning at high RPM (typically 12,000 to 24,000 RPM) mounted in gimbal rings providing degrees of freedom:

- **Rigidity in Space (Gyroscopic Inertia):** The property of a spinning rotor to resist any force attempting to alter its plane of rotation in space (Newton's 1st Law). Directly proportional to: 1. Rotor Mass (m); 2. Radius of gyration (r); 3. Rotational Speed (omega): Angular Momentum $L = I \times \omega$.
- **Gyroscopic Precession:** When an external torque (force) is applied to deflect the spin axis, the rotor resists in the direction of force and instead tilts at a point 90° LATER in the direction of rotation! Precession rate is inversely proportional to rotor speed.

2. CLASSIC GYROSCOPIC INSTRUMENTS: DI, AI & TURN COORDINATOR

Instrument	Rotor Spin Axis & Degrees of Freedom	Operating Function & Real/Apparent Errors
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Directional Gyro (Heading Indicator)	HORIZONTAL spin axis; 2 degrees of freedom (tied to azimuth).	Indicates aircraft heading. Free gyro: suffers Apparent Drift due to Earth's rotation: $\text{Drift} = 15^\circ/\text{hour} \times \sin(\text{Latitude})$. At 60°N: $15 \times \sin(60^\circ) = 13^\circ/\text{hour}$ apparent drift. Must be reset to magnetic compass every 10-15 min!
Artificial Horizon (Attitude Indicator)	VERTICAL spin axis; 2 degrees of freedom (tied to vertical).	Indicates Pitch and Roll attitude. Erected by pendulous vane air jets or electrical torque motors. Acceleration errors (indicates false climb and right roll on take-off acceleration); Turning errors (indicates false climb and opposite bank on rolling out of 180° turn).
Turn & Slip Indicator (Turn Coordinator)	HORIZONTAL spin axis; ONE degree of freedom with calibrated restraining springs.	Rate gyro: rate of yaw precesses gimbal against spring, deflecting needle. Rate 1 turn = 3°/sec (360° turn in 2 minutes). Slip ball (inclinometer) measures apparent gravity/centrifugal balance (shows slip or skid).

[!] EXAM TRAP (AVIATIONEXAM) - Apparent Drift (Earth Rate Wander) Calculation Formula

- Formula: $\text{Apparent Wander} = 15^\circ \times \sin(\text{Latitude})$ per hour! - At the Equator (Lat 0°): $\text{Drift} = 15 \times \sin(0^\circ) = \text{ZERO drift!}$ - At the North Pole (Lat 90°): $\text{Drift} = 15 \times \sin(90^\circ) = 15^\circ$ PER HOUR clockwise! - Direction: In the Northern Hemisphere, a free directional gyro drifts to the RIGHT (clockwise); in the Southern Hemisphere, it drifts to the LEFT!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Gyro Earth Rate Apparent Drift**

Formula: Drift = $15 \text{ deg} \times \sin(\text{Latitude})$ per hour. Example at 60 N: $\sin(60) = 0.866 \rightarrow$

Drift = $15 \times 0.866 = 13 \text{ deg/h}$ to the right. At Equator (0 deg): $\sin(0) = 0 \rightarrow$ Drift is

ZERO. At Poles (90 deg): $\sin(90) = 1.0 \rightarrow$ Drift is MAXIMUM (15 deg/h).

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: A free directional gyroscope is operated at Latitude 60° North. What is the calculated apparent drift rate due to Earth's rotation?

- [A]: 0° per hour
- [B]: 7.5° per hour
- [C]: 13.0° per hour
- [D]: 15.0° per hour

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Formula: Earth Rate Drift = $15^\circ \times \sin(\text{Lat})$ per hour = $15^\circ \times \sin(60^\circ) = 15 \times 0.866 = 12.99^\circ$ per hour ($\sim 13.0^\circ/\text{hr}$).

Q2: In a basic air-driven artificial horizon (attitude indicator), what false indication occurs when the aircraft accelerates forward?

- [A]: A false nose-down pitch
- [B]: A false pitch-UP and a slight roll to the RIGHT
- [C]: A false rapid turn to the left
- [D]: No error occurs

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Take-off forward acceleration closes the rear pendulous erection vanes under inertia, causing the gyro to precess, generating a false indication of pitch-up and a slight false bank to the right.

Q3: How many degrees of freedom does the rate gyro inside a Turn and Slip indicator possess?

- [A]: One degree of freedom (gimbal constrained by springs)
- [B]: Two degrees of freedom
- [C]: Three degrees of freedom
- [D]: Zero degrees of freedom

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. A rate gyro has only ONE gimbal ring (one degree of freedom), restrained by calibrated mechanical centering springs. Gyro precession directly measures the rate of angular yaw.